

KAUNAS, Lithuania

WMO: 266290

Lat: 54.8834N

Lon: 23.8349E

Elev: 77

StdP: 100.40

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.7	-15.5	-21.2	0.6	-18.5	-18.1	0.8	-15.1	10.8	-0.6	10.0	-1.6	1.9	160	0.570

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.4	28.6	19.9	26.8	18.9	25.1	18.0	21.1	26.4	20.0	24.8	19.0	23.6	3.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.3	14.2	23.5	18.3	13.3	22.2	17.3	12.4	21.3	61.4	26.2	57.5	24.7	54.3	23.6	25.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.1	7.2	DB	-21.8	31.4	4.0	1.8	-24.7	32.7	-27.0	33.8	-29.2	34.8	-32.1	36.2	(4)
(5)			WB	-22.0	22.8	3.9	1.2	-24.8	23.7	-27.1	24.4	-29.3	25.1	-32.1	26.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.5	-3.6	-2.8	1.1	7.7	12.8	15.9	18.5	17.8	13.1	7.4	2.7	-1.5	(6)	
	DBStd	8.89	5.88	5.70	4.34	4.37	4.02	3.22	3.10	2.94	3.26	3.72	4.11	5.33	(7)	
	HDD10.0	1851	422	358	276	95	17	1	0	0	9	98	219	356	(8)	
	HDD18.3	4053	681	591	534	318	178	86	37	46	159	339	469	614	(9)	
	CDD10.0	932	0	0	1	27	102	178	262	243	103	17	0	0	(10)	
	CDD18.3	93	0	0	0	1	5	14	42	30	2	0	0	0	(11)	
	CDH23.3	798	0	0	0	6	54	122	343	253	19	0	0	0	(12)	
	CDH26.7	167	0	0	0	0	6	18	84	58	2	0	0	0	(13)	
(14) Wind	WSAvg	3.5	4.2	4.1	3.9	3.5	3.2	3.1	2.8	2.8	3.1	3.7	4.1	4.2	(14)	
(15) Precipitation	PrecAvg	631	40	31	36	39	53	69	81	75	55	55	51	47	(15)	
	PrecMax	845	92	68	81	86	152	138	200	174	111	167	121	77	(16)	
	PrecMin	446	3	4	9	4	11	7	6	5	8	4	11	16	(17)	
	PrecStd	91	20	14	18	20	27	30	45	42	26	35	22	16	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	7.7	14.5	24.1	27.4	29.0	31.6	31.1	25.8	18.4	11.9	8.9	(19)	
		MCWB	6.5	6.0	8.6	14.9	18.4	20.0	20.8	20.7	17.7	14.1	10.3	7.8	(20)	
	2%	DB	5.6	6.2	11.2	20.8	24.8	26.6	29.0	28.1	22.6	15.9	10.2	7.1	(21)	
		MCWB	4.5	4.8	7.0	12.9	16.5	18.7	20.4	19.2	16.4	13.0	9.0	6.2	(22)	
	5%	DB	4.2	4.7	8.8	17.9	22.8	24.6	27.0	26.2	20.3	14.1	8.9	5.8	(23)	
		MCWB	3.3	3.6	5.9	11.2	15.5	17.7	19.6	18.3	15.4	12.0	7.9	5.0	(24)	
	10%	DB	2.8	3.6	6.7	15.4	20.7	22.7	25.1	24.2	18.6	12.7	7.9	4.5	(25)	
		MCWB	2.3	2.7	4.6	10.0	14.5	16.7	18.8	17.5	14.6	11.0	7.1	3.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.6	6.4	9.3	15.6	19.8	21.5	23.2	23.0	18.6	15.0	10.6	8.1	(27)	
		MCDB	7.4	7.3	12.9	23.4	25.9	27.0	28.7	28.6	23.8	17.2	11.4	8.7	(28)	
	2%	WB	4.8	4.9	7.5	13.5	17.8	19.7	21.5	20.6	17.2	13.5	9.2	6.3	(29)	
		MCDB	5.5	6.0	10.3	19.2	23.0	25.1	26.7	25.5	20.9	15.3	10.0	6.9	(30)	
	5%	WB	3.4	3.8	6.1	11.9	16.5	18.5	20.4	19.4	16.1	12.3	8.1	5.1	(31)	
		MCDB	4.0	4.6	8.4	16.8	21.4	23.2	25.3	24.1	19.3	13.8	8.8	5.8	(32)	
	10%	WB	2.3	2.7	4.8	10.5	15.1	17.3	19.4	18.4	15.1	11.2	7.1	3.8	(33)	
		MCDB	2.9	3.5	6.5	14.5	19.2	21.6	23.7	22.7	17.9	12.6	7.8	4.4	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.8	6.2	9.3	10.2	9.4	9.4	9.7	8.3	6.0	3.6	3.7	(35)	
		MCDBR	3.8	4.6	9.6	13.5	13.4	12.5	12.6	13.3	11.4	8.1	4.7	3.8	(36)	
	5% WB	MCWBR	3.6	3.8	6.3	6.8	6.5	6.0	5.3	5.5	5.9	5.3	4.0	3.5	(37)	
		MCWBR	3.6	4.4	8.6	12.3	11.7	10.8	10.8	10.8	9.7	7.2	4.4	3.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.298	0.330	0.366	0.408	0.396	0.388	0.425	0.419	0.382	0.357	0.338	0.299	(40)	
		taud	2.198	2.181	2.212	2.196	2.293	2.343	2.261	2.273	2.349	2.398	2.282	2.235	(41)	
	Ebn at Noon		620	730	794	811	847	858	815	793	775	707	568	533	(42)	
		Edh at Noon	61	88	108	126	121	117	125	116	94	72	57	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.57	1.32	2.60	4.08	5.33	5.55	5.17	4.33	3.01	1.48	0.54	0.36	(44)	
	RadStd		0.11	0.25	0.40	0.53	0.52	0.46	0.51	0.42	0.34	0.28	0.07	0.05	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.43	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (18 neighbors)		+0.42	N/A	N/A	N/A	N/A	N/A	N/A	-148	N/A	N/A

Nomenclature: See separate page